Question	Answer	Marks	Partial Marks
6(c)	258 or 257.9 to 258.0…	**5**	**B4** for 67.9 to 68.0… OR **M2** for [cos =] $\left(\frac{11^2+13.7^2-14^2}{2\times13.7\times11}\right)$ **A1** for 0.3738 to 0.376 or **M1** for $14^2 = 11^2 + 13.7^2 - 2\times 11 \times 13.7 \times \cos B$ **M1dep** on at least M1 for 190 + *their* angle *B*
6(d)(i)	2 44 pm or 14 44 cao	**4**	**B3** for 1 hour 44 or 1 hour 43.6 to 1 hour 43.8 or 104 or 103.6 to 103.8 or **B2** for 1.727 to 1.73 or **M2** for $\frac{32}{10\times1.852}\times60$ or **M1** for 32 ÷ (10 × 1.852)
6(d)(ii)	7.857 to 7.88	**3**	**M2** for $\frac{x}{13.7}=\cos55$ oe or **M1** for dist to *H* occurs when perpendicular from *B* meets *CH* soi
7(a)(i)	10 100	**3**	**M2** for $30\times70+2\times30\times40+2\times40\times70$ or **M1** for 30 × 40 or 30 × 70 or 40 × 70
7(a)(ii)	16	**3**	**M2** for 2 fit width, 2 fit height and 4 fit length soi or **M1** for 70, 30 or 40 ÷ 15 or 20
7(b)(i)	$\frac{1}{3}\pi r^2\times3r$ = *their* (750 ÷ 8.9) oe	**M2**	**M1** for using 750 and 8.9 correctly in $v=m/d$ oe or 750 ÷ 8.9
	$r^3=\frac{their(750\div8.9)}{\pi}$ oe	**M1dep**	
	$r = 2.993\ldots$	**A1**	

Question	Answer	Marks	Partial Marks
7(b)(ii)	117 or 116.9 to 117.2	**5**	**M4** for $\pi \times 2.99^2 + \pi \times 2.99 \times \sqrt{2.99^2 + (3 \times 2.99)^2}$ oe or **M3** for $\pi \times 2.99 \times \sqrt{2.99^2 + (3 \times 2.99)^2}$ or **M2** for $\sqrt{2.99^2 + (3 \times 2.99)^2}$ or **M1** for $2.99^2 + (3 \times 2.99)^2$ or for $\pi \times 2.99^2$
8(a)	Correct sketch with roots indicated at $x = -9$ and $x = 2$ and y intercept = –18 Minimum should be in 3rd quadrant	**4**	**B1** for U shaped parabola **B2** for roots at –9 and 2 on diagram or **M1** for $(x + 9)(x - 2)$ [= 0] **B1** for y – intercept at –18 on diagram Maximum 3 marks if sketch not fully correct
8(b)(i)	$2x - 3$	**2**	**B1** for $2x + k$ or $kx^{[p]} - 3$
8(b)(ii)	(1.5, –30.25) oe	**3**	**B2** for $x = 1.5$ or **M1** for *their* **(b)(i)** = 0 or for $(x - 1.5)^2$
8(c)	$x^2 - x - 33$ [= 0] seen	**B1**	
	$\frac{[--]1 \pm \sqrt{([-]1)^2 - 4(1)(-33)}}{2 \times 1}$ oe	**B2FT**	**FT** *their* quadratic dep on no factors **B1** for $\sqrt{([-]1)^2 - 4(1)(-33)}$ or better or **B1** for $\frac{[--]1 + \sqrt{q}}{2(1)}$ oe or $\frac{[--]1 - \sqrt{q}}{2(1)}$ oe
	–5.27 or –5.267 to –5.266 **and** 6.27 or 6.266 to 6.267	**B2**	**B1** for each If 0 scored, **SC1** for –6.27 and 5.27
	(–5.27, 15.53 or 15.54) **and** (6.27, –7.53 or – 7.54)	**B1**	
9(a)(i)	13	**1**	
9(a)(ii)	–20	**1**	**FT** 6 – 2(*their* **(a)(i)**)